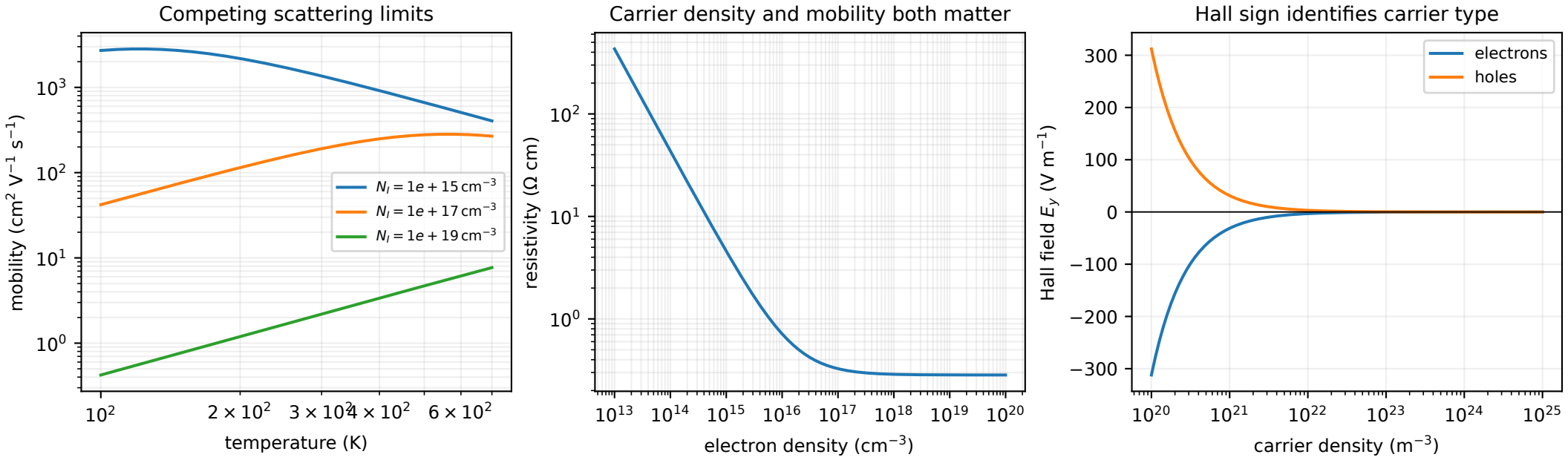
Competing scattering limits
mobility (cm² V⁻¹ s⁻¹)
temperature (K)
$N_I = 1e+15\,\mathrm{cm}^{-3}$
$N_I = 1e+17\,\mathrm{cm}^{-3}$
$N_I = 1e+19\,\mathrm{cm}^{-3}$
Carrier density and mobility both matter
resistivity (Ω cm)
electron density (cm⁻³)
Hall sign identifies carrier type
Hall field E_y (V m⁻¹)
carrier density (m⁻³)
electrons
holes